

Aneesh Krishna

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EDUCATION

Purdue University

May 2027

B.S. Computer Science & Statistics

Minors: Math & Finance

- **Orgs/Awards:** USACO Silver · 6x Hackathon Winner · Boiler Blockchain · Data Mine

EXPERIENCE

Amazon | *Systems Development Intern*

May 2026 – Aug 2026

- Built a FastAPI automation service on AWS, cutting tier-generation time by **98%** across **100 MB** Excel workbooks
- Orchestrated a **6-stage** async data processing pipeline, scaling backend throughput by **20x** across **10,000+ jobs**
- Developed a low-level XML ZIP manipulation layer, ensuring **100% byte-fidelity** for **5,000 corporate assets**
- Automated live distributed data enrichment from Redshift, eliminating **12,000** manual reconciliation steps per month

Caterpillar | *Machine Learning Intern*

Aug 2025 – Dec 2025

- Architected a PyTorch multi-horizon transformer, **reducing WAPE** by **24%** across **1M+ records**
- Scaled distributed ML inference via Kubernetes, slashing forecasting latency for **500+ SKUs** down to **2.8 minutes**
- Deployed an automated Azure MLOps pipeline, processing **100K+ telemetry events/day** for real-time model serving

Stealth Startup | *Software Engineer*

Jun 2025 – Aug 2025

- Implemented an Auth0–Stripe payment gateway with PaymentIntent state management, handling **20,000 checkouts**
- **Tripled API throughput** by replacing client polling with server-issued session tokens, reducing request load by **70%**

Playtoon-It | *Software Developer*

Mar 2025 – May 2025

- Optimized game state synchronization to lower **server costs** by **30%** and **packet loss** by **25%** (**30K player sessions**)
- Overhauled ECS via event-driven architecture and object pooling, cutting **latency spikes** by **30%** and **churn** by **10%**

PROJECTS

Cathode – CPU Graphics Engine

- Built a CPU rasterizer (AArch64 NEON SIMD), rendering **50+ 1440p scenes** at **<2ms/frame** with **4x speedup**
- Designed a NTSC DSP pipeline with I/Q modulation, sustaining **114 MS/s** at **<0.5% NRSME** across **13K tests**

POSIX – Custom UNIX Shell

- Built an AST-based POSIX shell with pipes, redirection & job control, passing **1,200+ tests** across **45 shell features**
- Engineered process groups, signals & TTY control for **500 jobs**, passing **180K stress iterations** with zero deadlocks

Vibe – Social Network

- Built a WebSocket gateway with consistent hashing, sustaining **130K concurrent clients** at **<40ms p95** across **8 nodes**
- Engineered ordered fanout with backpressure & idempotent delivery, sustaining **250K msg/s** with **<0.01% loss**

SKILLS

- **Languages:** Python · TypeScript · C/C++ · Go · Rust · Java · SQL · R · C#
- **Frameworks:** Next.js · LangChain · PyTorch · TensorFlow · Redis · Kafka · RabbitMQ · Valgrind
- **Tools:** Git · Docker · MongoDB · Firebase · Linux · Terraform · Kubernetes · Prometheus · Grafana · gPRC